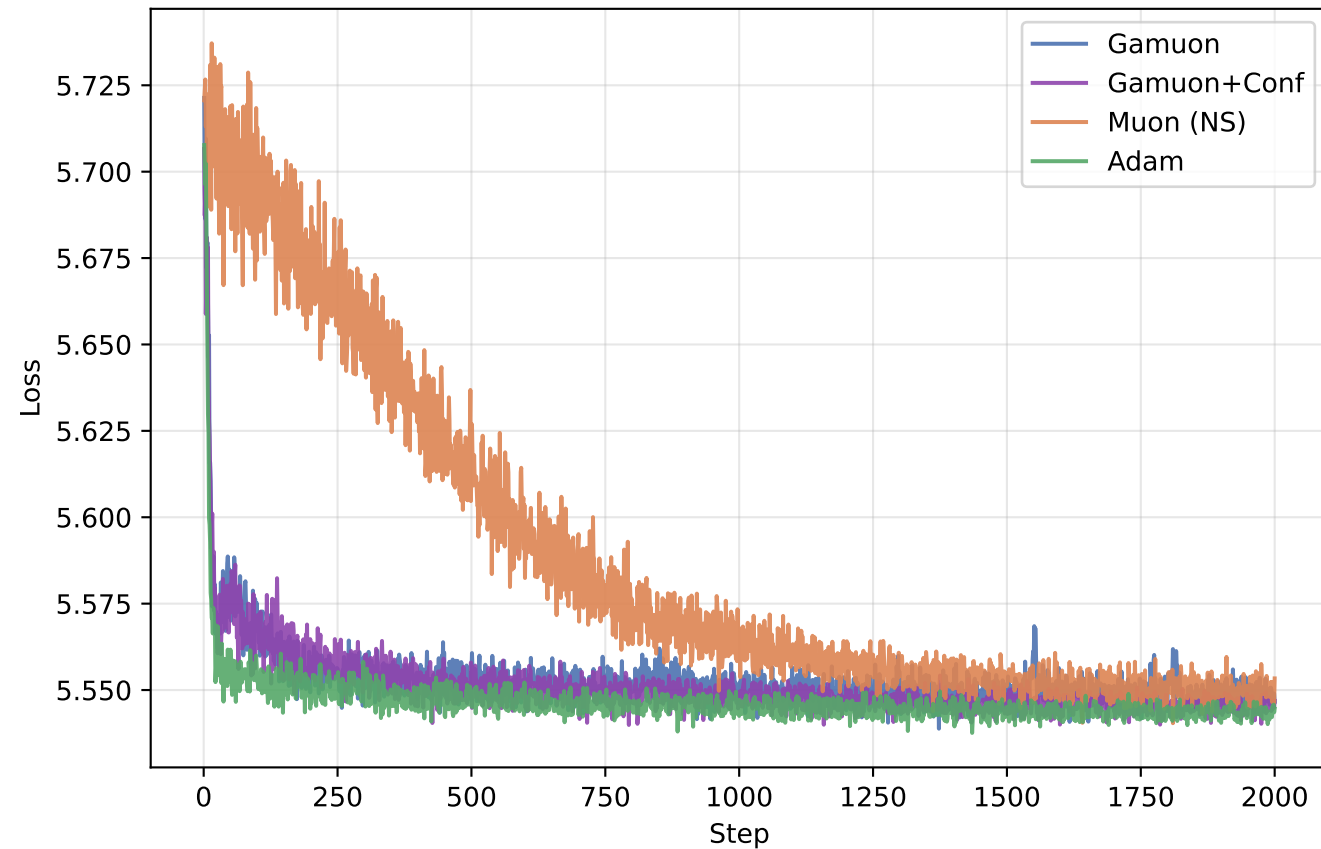
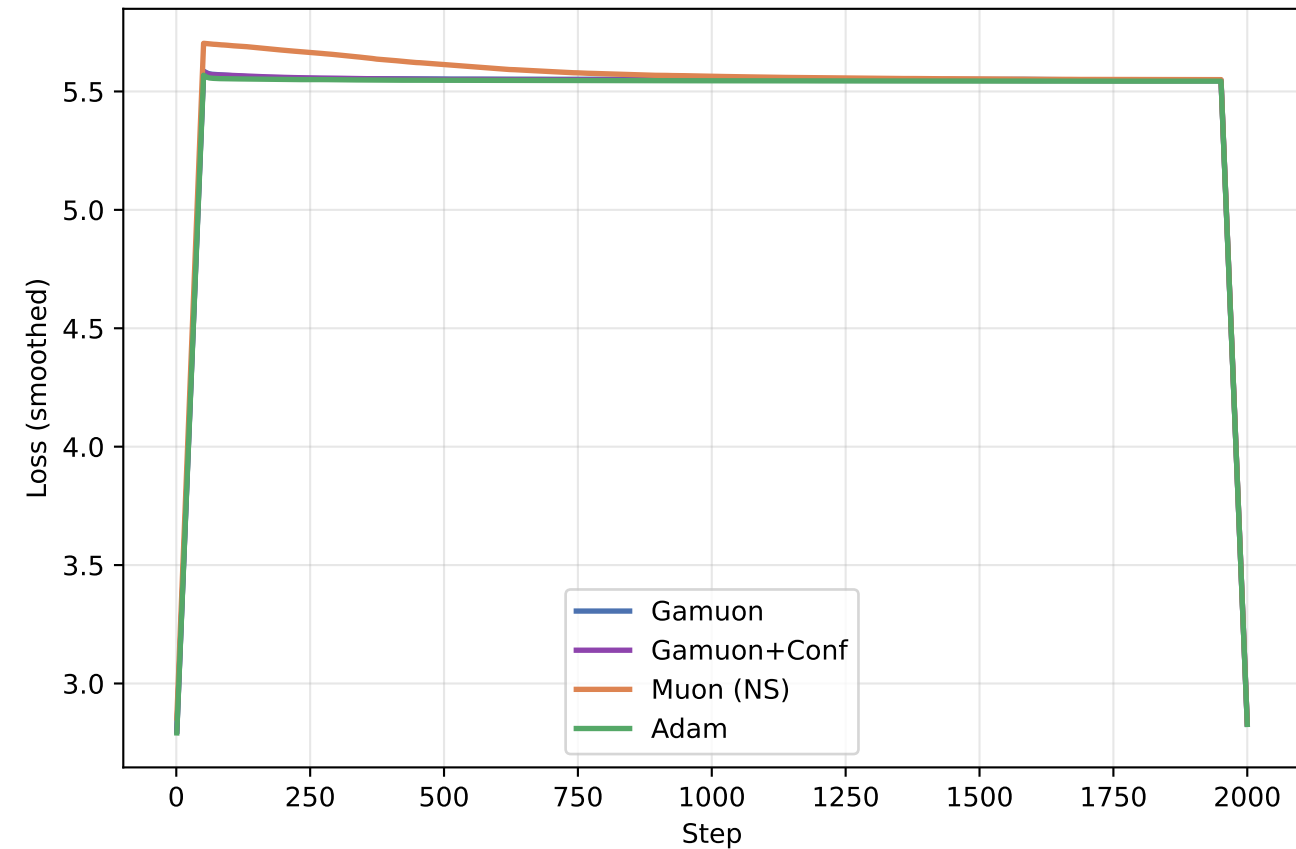


# Optimizer Comparison — Small Transformer

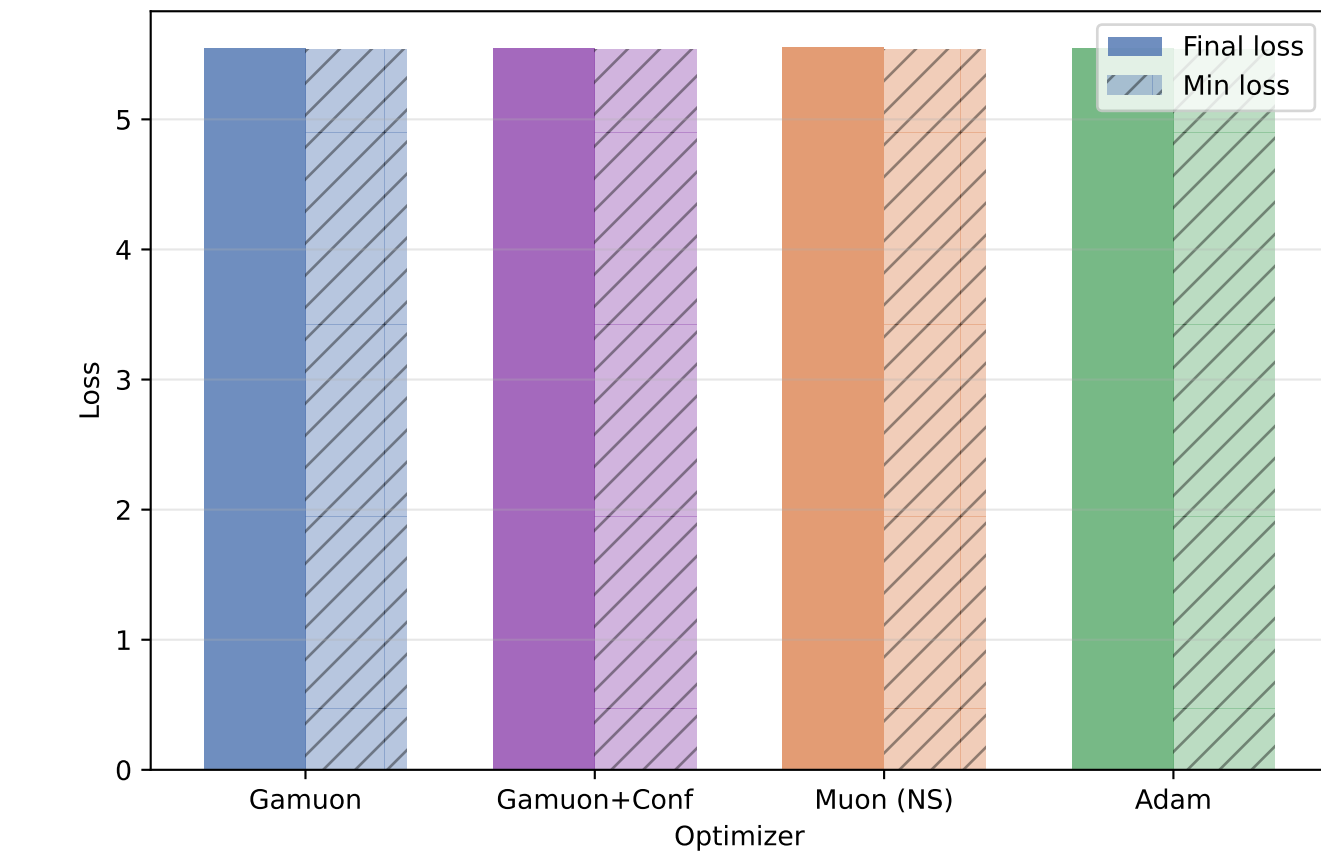
## Training Loss vs Steps



## Smoothed Loss (window=100)



## Final & Minimum Loss



## Per-Step Wall-Clock Time

